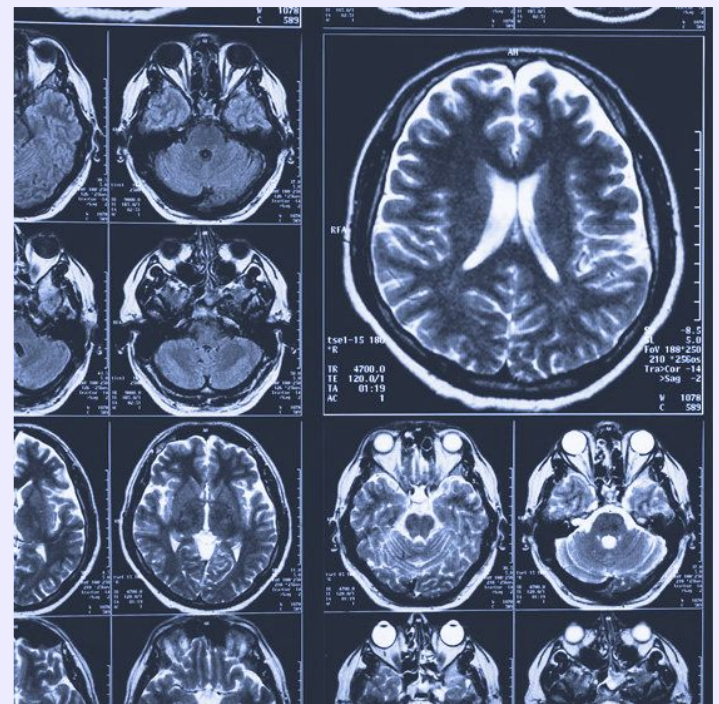


Alzheimer's Progression Digital Twin

**Predicting Brain Atrophy Using Machine
Learning on MRI Data**

By: Evani Menon, Manojna Reddy Kamaram, Diksha
Kaushik, Palak, Ishaan Arora, Isobel Kuriyan

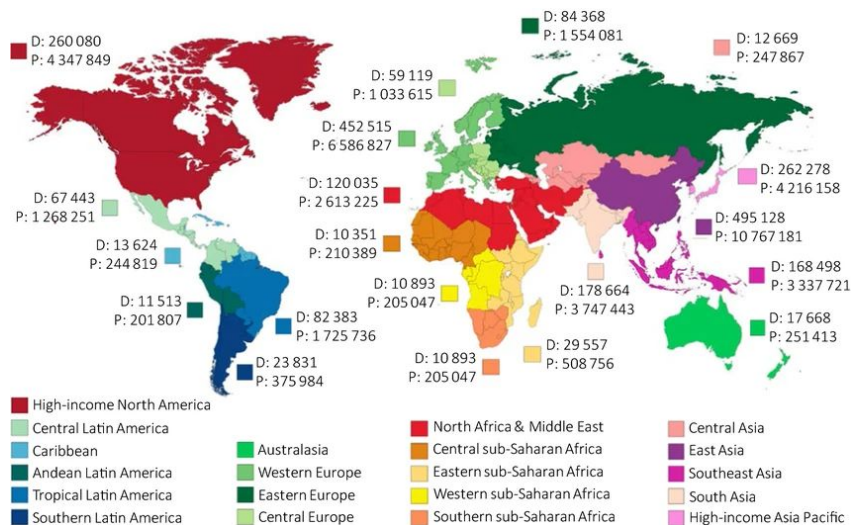


- **50-90%** of Alzheimer's cases remain **undiagnosed**
- Disease **progresses silently** for **10-20 years** before symptoms
- Current tools **cannot predict individual patient trajectories**
- By the time diagnosis occurs, interventions are less effective

Why This Matters:

- Over 55 million people worldwide have dementia, **60-70%** being **Alzheimer's** (140M globally by 2050)
- Early diagnosis could reduce number of cases and treatment costs
- Need: **Non-invasive tool to predict who will decline and when**

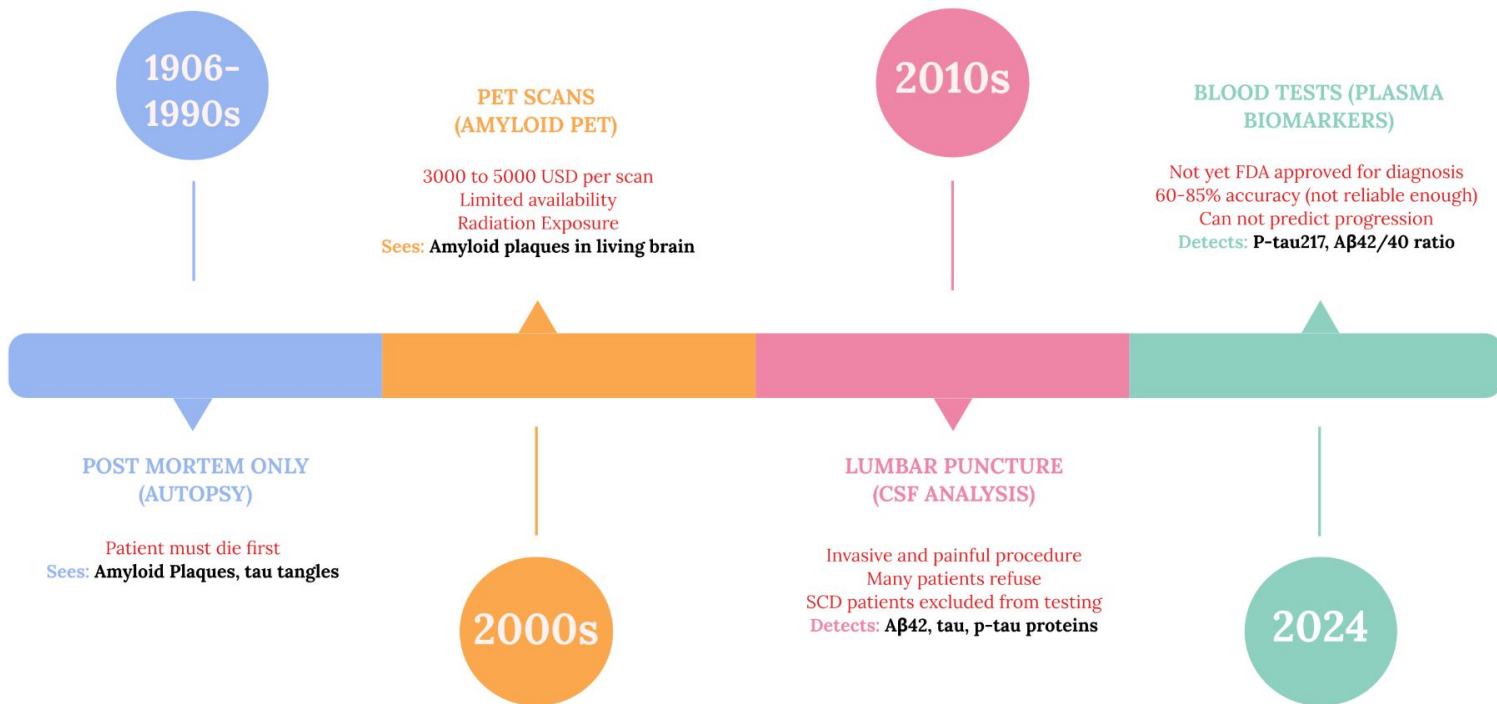
The Alzheimer's Crisis



Global deaths (D) and prevalence (P) for Alzheimer's disease and other dementias in 2016. Data are n (95% UI). UI = uncertainty interval.

Data extracted from the Global Burden of Disease Study of Alzheimer's disease and other dementias 2016

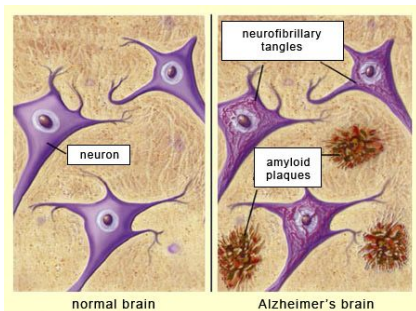
The Evolution of Alzheimer's Diagnosis:



What these Tests Look for:

Amyloid- β Plaques

- Sticky protein clumps between neurons
- Disrupt cell communication
- Detected by: PET scan, CSF, blood test

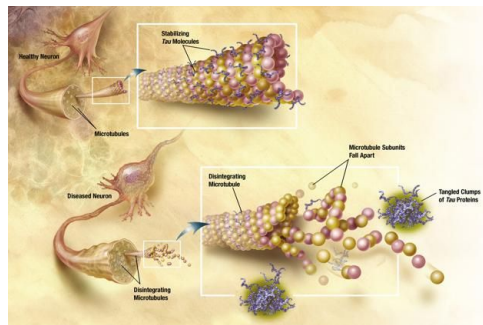


Metabolic Changes

- Reduced glucose use in brain
- Neurons can't produce energy
- Detected by: FDG-PET scan

Tau Tangles (Neurofibrillary Tangles)

- Twisted protein fibers inside neurons that kill brain cells
- Detected by: CSF, blood test (p-tau)

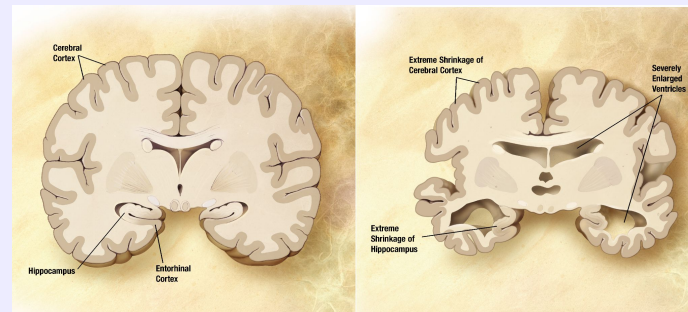


Neuroinflammation

- Brain's immune response
- Causes further damage
- Detected by: PET scan, CSF

Brain Atrophy (Shrinkage)

- Hippocampus shrinks first (memory center)
- Ventricles expand (fluid-filled spaces)
- Cortex thins (outer brain layer)
- Detected by: MRI, PET scan



Digital Twin Concept

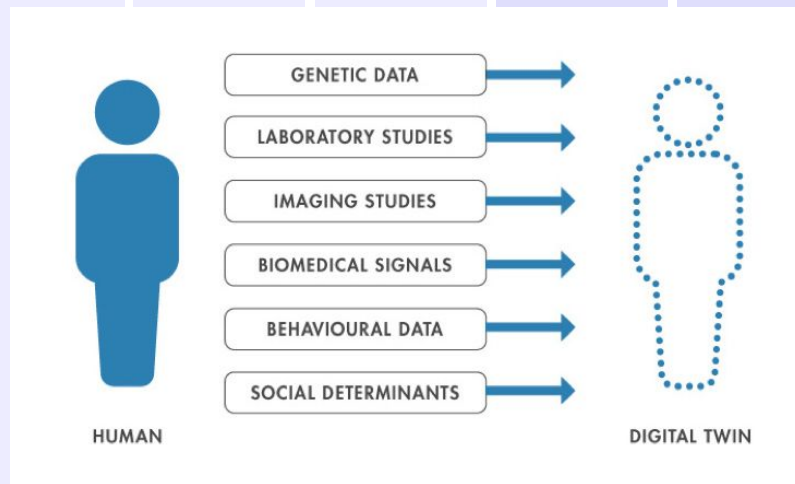
A Digital Twin is a virtual representation that predicts how a real-world system will change over time

We're building a machine learning model that:

- Analyzes brain MRI scans + clinical data
- Learns from historical patterns and predicts future disease progression (personalized prediction)
- Forecasts cognitive decline 12-24 months ahead
- Visualizes personal model of atrophy patterns in patient's brain
- Simulates "what-if" scenarios

The gap our project fills:

- Non-invasive (just MRI- already standard)
- Predicts future progression (not just diagnosis)
- Shows WHERE brain will atrophy
- Accessible and affordable



Digital Twins in Alzheimer's Research

PREVIOUS WORK

RECENT BREAKTHROUGH (January 2025) Amato et al.

(Proves digital twins work for Alzheimer's and why our project is relevant)

- **EEG-based Digital Twin** (<https://pmc.ncbi.nlm.nih.gov/articles/PMC12125947/>)
- Uses brainwave recordings (EEG)
- Predicts CSF biomarkers
- Functional brain activity analysis

OUR WORK AND HOW IT'S DIFFERENT

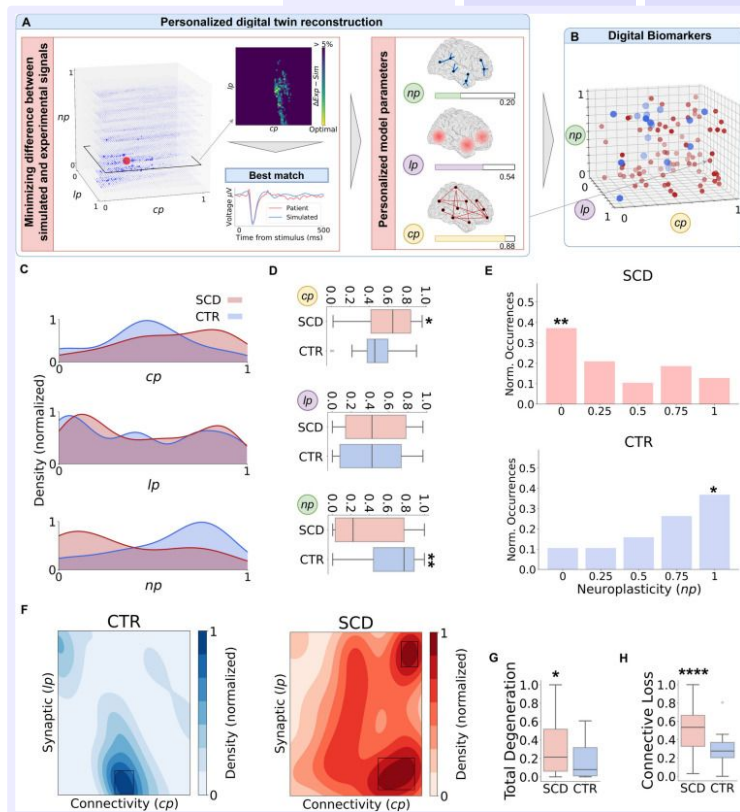
MRI-based Structural Digital Twin

- Uses brain imaging (MRI)
- Predicts brain atrophy patterns
- Structural anatomy analysis

Complementary approach - predicts **WHERE** brain will shrink

COMBINED FUTURE

Recent work showed EEG-based twins can predict disease markers. We're tackling the structural side - predicting actual brain anatomy changes. These approaches complement each other.



Data Sources

PRIMARY DATASET: ADNI

(Alzheimer's Disease Neuroimaging Initiative)

WHAT WE ARE USING:

1) **MRI Scans (ADNI1 Collection)**

Type: T1-weighted structural brain imaging

Format: NIfTI (.nii.gz)

Subjects: 200-300 patients (our target subjects out of ~5k) and 5 demo patients selected

Timepoints: Baseline, 12mo, 24mo

Size: ~50 GB (for 200-300 subjects x 3 timepoints)

2) **Clinical Data (ADNIMERGE.csv)**

MMSE cognitive test scores

Demographics (age, gender, education)

APOE4 genetic risk factors

Brain volume measurements

Diagnosis labels (CN, MCI, AD)

DATA ACCESS:

Access approved for ADNI usage

Download via ADNI portal: adni.loni.usc.edu

All data is de-identified (HIPAA compliant)

RID	VISCODE	MMSE	DX	Hippocampus	Gender	Education	APOE4	visit_num
4.0	m12	26.0	MCI	6166.0	Male	10.0	0.0	2.0
4.0	m18	27.0	MCI	6057.0	Male	10.0	0.0	3.0
4.0	m36	25.0	MCI	5986.0	Male	10.0	0.0	5.0
5.0	bl	29.0	CN	6546.0	Male	16.0	0.0	0.0
5.0	m06	29.0	CN	6356.0	Male	16.0	0.0	1.0
5.0	m12	30.0	CN	6454.0	Male	16.0	0.0	2.0
5.0	m24	29.0	CN	6203.0	Male	16.0	0.0	4.0
5.0	m36	30.0	CN	6038.0	Male	16.0	0.0	5.0
6.0	bl	25.0	MCI	5004.0	Female	13.0	0.0	0.0
6.0	m06	21.0	MCI	4995.0	Female	13.0	0.0	1.0
6.0	m12	26.0	MCI	4930.0	Female	13.0	0.0	2.0
6.0	m18	24.0	MCI	4831.0	Female	13.0	0.0	3.0
6.0	m24	26.0	MCI		Female	13.0	0.0	4.0
6.0	m36	22.0	MCI	4627.0	Female	13.0	0.0	5.0
7.0	bl	20.0	Dementia	6357.0	Male	10.0	1.0	0.0
7.0	m06	23.0	Dementia	6397.0	Male	10.0	1.0	1.0
7.0	m12	20.0	Dementia		Male	10.0	1.0	2.0
7.0	m24	20.0	Dementia	5889.0	Male	10.0	1.0	4.0
8.0	bl	28.0	CN	5511.0	Female	18.0	0.0	0.0
8.0	m06	29.0	CN	5549.0	Female	18.0	0.0	1.0
8.0	m12	28.0	CN		Female	18.0	0.0	2.0
8.0	m36	29.0	CN		Female	18.0	0.0	5.0
8.0	m48	30.0	CN		Female	18.0	0.0	6.0
8.0	m60	28.0	CN		Female	18.0	0.0	7.0

Hippocampus values available for ~500 ADNI1 subjects"

Understanding our Data

MMSE score

Mini Mental State Exam. A 30-point cognitive test. A patient scoring 28 today might score 22 in a year.

This is the **primary prediction target** that the LSTM is trying to forecast.

Diagnosis (CN/MCI/Dementia)

CN = Cognitively Normal

MCI = Mild Cognitive Impairment

Dementia = Alzheimer's

This is used for classification tasks and for picking demo patients

Hippocampus Volume

The hippocampus is the first brain region to physically shrink in Alzheimer's. A healthy adult has $\sim 6,000\text{mm}^3$ total. AD patients can drop to $\sim 4,000\text{mm}^3$.

This is the biomarker that we will **visualize shrinking in 3D**.

It's also a powerful model feature because physical atrophy often predicts cognitive decline before the patient even notices symptoms.

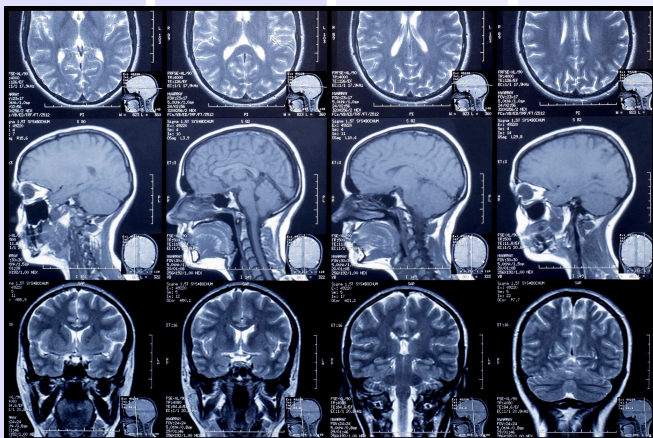
Gender & Education

Both affect baseline cognitive scores. Highly educated patients often "mask" early decline better, so their MMSE stays artificially high longer. Gender affects disease progression rate. Without these, the model could make systematically wrong predictions for certain subgroups.

APOE4

The strongest known **genetic risk factor** for Alzheimer's. Carrying one copy of the $\epsilon 4$ allele doubles your risk; two copies increases it 8–12x. It's a 0/1/2 number per patient that never changes. Including it lets our model adjust trajectory predictions based on genetic risk. A patient with APOE4=2 and MCI is very likely to convert to dementia; APOE4=0 is much less likely.

Project Impact



CLINICAL IMPACT:

- Early Detection Window: Predict 12-24 months ahead
- Enables intervention when treatments work best
- Could reduce AD cases

Non-Invasive Alternative to Lumbar Puncture

- 67% of clinics refuse CSF testing for SCD patients
- Our approach: Just MRI (already standard imaging)

Personalized Precision Medicine

- Not "you have Alzheimer's" but "here's YOUR timeline"
- Visualize WHERE your brain will atrophy
- Enables targeted intervention strategies

Cost-Effective Screening

- MRI: \$400-600 (one-time, already done for diagnosis)
- PET scan: \$3,000-5,000 (not widely available)
- CSF test: \$500-1,000 + painful procedure

Healthy controls		Mild Cognitive Impairment							
		Typical AD	Limbic-predominant	Hippocampal-sparing	Minimal atrophy				
A									
P									
R									
L									
MTA = Right 0 Left 0 PA = 0 GCA-F = 0		MTA = Right 4 Left 4 PA = 1 GCA-F = 1		MTA = Right 4 Left 4 PA = 0 GCA-F = 0		MTA = Right 0 Left 0 PA = 2 GCA-F = 1		MTA = Right 0 Left 0 PA = 0 GCA-F = 0	

Project Limitations

Clinical scope

- Predicts progression only; cannot diagnose
- Requires clinical diagnosis as prerequisite
- Supportive tool; does not replace physician judgment

Prediction accuracy

- 75–80% accuracy; 20–25% error rate
- Cannot predict acute events (stroke, injury)
- High individual variability in disease trajectory

Imaging dependency

- Requires baseline structural MRI
- Contraindicated: pacemakers, metal implants
- MRI cost remains \$400–600 per scan

Dataset limitations

- Training sample: 4,722 subjects, 2,475 sequences
- ADNI cohort: predominantly white, educated, high-income
- Generalizability to diverse populations unvalidated

Molecular blindspot

- Does not detect amyloid plaques or tau tangles directly
- Structural atrophy is downstream of molecular pathology
- Damage may be established before MRI changes appear
- CSF/PET biomarkers still required for confirmation

Model interpretability

- Deep learning model — black box architecture
- Cannot explain mechanistic basis of predictions
- Insufficient explainability for clinical acceptance

Computational constraints

- GPU required for training (high cost)
- Training time: 6–8 hours per run
- Inference not real-time at scale

Disease heterogeneity

- MRI atrophy pattern shared across dementia subtypes
- Cannot reliably distinguish from vascular or frontotemporal dementia

Methodology and Approach

Feature Extraction (3D ResNet)

- Pre-trained Med3D model (transfer learning)
- Extracts 512 features from each MRI scan
- No building from scratch - using proven architecture

Temporal Prediction (LSTM)

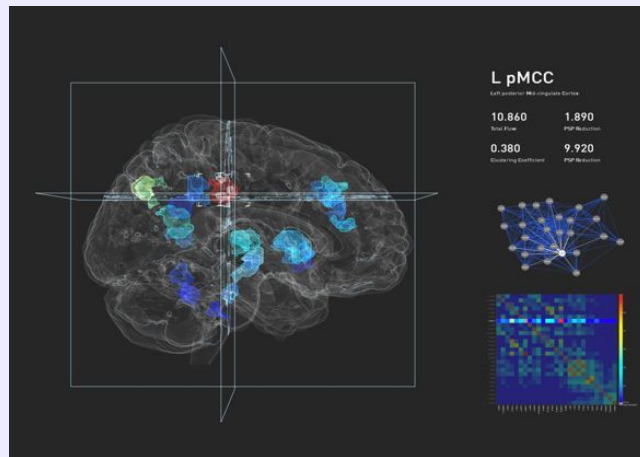
- Learns progression patterns over time
- Predicts brain changes 12-24 months ahead
- Multi-task: cognitive scores + brain volumes

Visualisation

- 3D brain rendering (Nilearn, PyVista)
- Atrophy heat maps showing where brain shrinks
- Interactive comparisons: baseline → predicted

Tools

Python, PyTorch, ADNI data, Nilearn, Local Python environment + Google Colab GPU (for LSTM training)



Sample Brain Visualisation
(not exact)

Project Timeline & Deliverables — Alzheimer's Digital Twin

	WEEK 1	WEEK 2	WEEK 3	WEEK 4
PHASE	Data preprocessing	Feature extraction	LSTM training + prediction	Visualization + evaluation
METHOD / TOOL	FSL / FreeSurfer skull-strip, normalize	3D ResNet on MRI volumes	LSTM network sequence learning	Matplotlib / Nilearn metric computation
INPUT	Raw MRI scans + clinical data (ADNI)	Preprocessed MRI + tabular features	Extracted feature vectors per time point	Model predictions + ground truth labels
OUTPUT / DETECTS	Clean, aligned dataset train / val / test split	Regional atrophy maps + feature embeddings	MMSE / CDR score progression predictions	3D brain renders predicted vs. actual plots
LIMITATION	Small dataset (~200–300 pts) ADNI bias	No molecular markers structural changes only	Black box model overfitting risk	75–80% accuracy not clinically validated
DELIVERABLE	Preprocessed dataset	Saved ResNet weights	Trained LSTM model	Final report + code repo

Feasibility

- Proven Technology: ResNet + LSTM are standard, well-tested
- Available Data: ADNI is accessible for academic use
- Manageable Scope: 200-300 subjects sufficient for course project
- Free Computing: Google Colab provides free GPU access
- 4-Week Timeline: Realistic with clear milestone

Evaluation metrics

Metric	Target	Why This Target?
Prediction	>0.70	Standard for regression
MMSE MAE	<2.5pts	Clinically meaningful
Classification Acc	75-80%	Better than blood tests
Progression AUC	>0.80	Distinguishes fast/slow

Contributions

Evani Menon

3D ResNet feature extraction
& model training

Manojna Kamaram

Clinical data integration &
analysis

Isobel Kuriyan

LSTM architecture & temporal
modeling

Diksha Kaushik

Data preprocessing & quality
control

Ishaan Arora

3D visualization & atrophy
mapping

Palak

Results evaluation &
documentation

Resources

Github

<https://github.com/evanimeno/alzheimers-digital-twin>

Data

ida.loni.usc.edu
(restricted, approved access)

Documentation

[Link to Google Docs](#)

Thank You!